

Rohan Disa

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Education

University of Southern California

Master of Science in Computer Science

Coursework: Analysis of Algorithms, Security Systems, Natural Language Processing, Distributed Systems

Jan 2025 – Dec 2026

Los Angeles, California

MIT World Peace University

Bachelor of Technology in Computer Science and Engineering

Coursework: Data Structures and Algorithms, Database Systems, Object Oriented Programming, Operating Systems

Dec 2020 – July 2024

Pune, India

Technical Skills

Languages & Databases: Python, Go, JavaScript, C++, SQL, PostgreSQL, Redis, MySQL, MongoDB

AI/ML Technologies: Keras, Tensorflow, NumPy, SciPy, MCP, Hugging-face, FAISS, Langchain, Langgraph

Development & OS: React, Node, Express, Git, Github, Redux, Docker, AWS, Linux, pytest, Cursor, Zapier

Experience

Ampere Computing

AI Software Engineer Intern

June 2026 – August 2026

Santa Clara, California

- **Reduced performance analysis time by 80%**, saving **30+ engineering hours per week**, by designing an agent and building its modular skill system that directs a model for hands-free, end-to-end root-cause analysis of performance bottlenecks
- Developed an asynchronous MCP server integrating 5 CLI tools plus performance utilities over SSH, enabling the agent to execute code across 1 runner and 6 SUTs (System Under Tests) with automated error handling
- Engineered an automated LLM-as-a-Judge evaluation framework over a 200-sample golden dataset, achieving **88% alignment with human expert benchmarks** (Cohen's Kappa = 0.81) to reliably evaluate multi-turn agent trajectories

USC Information Technology and Services

Software Engineer (part-time)

July 2025 – May 2026

Los Angeles, California

- Architected **Node.js** and **Express** backend services with **MongoDB** for persistent storage, integrating Shure APIs and building **RESTful endpoints** to manage device and classroom data, **reducing manual operational effort by 30+ hours/week**
- **Reduced backend memory usage from 16GB → 6GB** by addressing a long-lived HTTP client leak and added API pagination in the device-polling service, eliminating runtime failures and improving stability for long-running workloads
- Configured **CI/CD pipelines using GitHub Actions** and **Docker** to automate builds, testing, and deployments across environments, ensuring faster and more reliable releases

Amogh Data Innovations

Software Engineering Intern

July 2024 – Dec 2024

Pune, India

- Built video processing pipeline for warehouse automation system using **YOLO** for box detection and **GCP OCR** for multiple field extraction (invoice number, product name, etc.), **processing 150+ videos/day and cutting manual data entry 50%**
- Served the pipeline as a **Dockerized FastAPI service** with async batch inference, confidence-based validation with retry on low-quality frames, and **normalized PostgreSQL schemas** for queryable extraction history across **200+ unique products**

Open Source Contributions

- Fixed a state-merging bug in the Bedrock Converse API where initialization- and invocation-level configs collided, producing duplicate keys and schema violations ([langchain-aws #826](#))
- Deprecated Live Search in the ChatXAI integration with a backward-compatible migration path, preventing breakage against xAI's updated tools API ([langchain #34919](#))
- Optimized the HTMLSemanticPreservingSplitter by implementing a reversed iteration logic for element processing; ensured semantic structural integrity during document segmentation, critical for high-fidelity RAG pipelines ([langchain #34080](#))

Projects

Agent Eval Harness | Python, pytest, SciPy

April 2026 – May 2026

- Built a Python eval harness that reruns each agent task N times over a **32-task** suite with Wilson **95% CIs**
- Designed a three-tier grader with invariants, LLM-judge traces, and BH-corrected tests on **30+ tasks**
- Built the engine domain-agnostic with a plugin contract for task suites, so a new domain drops in without touching the core, shown with two unrelated suites through one loader

Capability Broker | Go, GitHub Actions

Feb 2026 – March 2026

- Built a Go broker that mints scoped, single-use credentials per tool call under deny-by-default
- Held a **100% block rate** on **10 red-team classes** at **0% FPR**, with ablations proving each control is required
- Enforced taint tracking, egress control, hash-chained audit, and mid-run revoke at **~170 ns** policy checks